HARD · GEOMETRY AND TRIGONOMETRY · ANSWER KEY

# Right triangles and trigonometry

04

3 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

## Q1 30

The correct answer is 30. In the figure given, since $\overline{BD}$ is parallel to $\overline{AE}$ and both segments are intersected by $\overline{CE}$, then angle BDC and angle AEC are corresponding angles and therefore congruent. Angle BCD and angle ACE are also congruent because they are the same angle. Triangle BCD and triangle ACE are similar because if two angles of one triangle are congruent to two angles of another triangle, the triangles are similar. Since triangle BCD and triangle ACE are similar, their corresponding sides are proportional. So in triangle BCD and triangle ACE, $\overline{BD}$ corresponds to $\overline{AE}$ and $\overline{CD}$ corresponds to $\overline{CE}$. Therefore, $\frac{BD}{CD} = \frac{AE}{CE}$. Since triangle BCD is a right triangle, the Pythagorean theorem can be used to give the value of CD: $6^2 + 8^2 = CD^2$. Taking the square root of each side gives $CD = 10$. Substituting the values in the proportion $\frac{BD}{CD} = \frac{AE}{CE}$ yields $\frac{6}{10} = \frac{18}{CE}$. Multiplying each side by CE, and then multiplying by $\frac{10}{6}$ yields $CE = 30$. Therefore, the length of $\overline{CE}$ is 30.

## Q2 113

The correct answer is 113. It's given that the legs of a right triangle have lengths 24 centimeters and 21 centimeters. In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two legs. It follows that if $h$ represents the length, in centimeters, of the hypotenuse of the right triangle, $h^2 = 24^2 + 21^2$. This equation is equivalent to $h^2 = 1{,}017$. Taking the square root of each side of this equation yields $h = \sqrt{1{,}017}$. This equation can be rewritten as $h = \sqrt{9 \cdot 113}$, or $h = \sqrt{9} \cdot \sqrt{113}$. This equation is equivalent to $h = 3\sqrt{113}$. It's given that the length of the triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$. It follows that the value of $d$ is 113.

## Q3

C

**C.** $58+58\sqrt{2}$

Choice C is correct. Since the triangle is an isosceles right triangle, the two sides that form the right angle must be the same length. Let $x$ be the length, in inches, of each of those sides. The Pythagorean theorem states that in a right triangle, $a^2+b^2=c^2$, where $c$ is the length of the hypotenuse and $a$ and $b$ are the lengths of the other two sides. Substituting $x$ for $a$, $x$ for $b$, and 58 for $c$ in this equation yields $x^2+x^2=58^2$, or $2x^2=58^2$. Dividing each side of this equation by 2 yields $x^2=\frac{58^2}{2}$, or $x^2=\frac{2\cdot 58^2}{4}$. Taking the square root of each side of this equation yields two solutions: $x=\frac{58\sqrt{2}}{2}$ and $x=-\frac{58\sqrt{2}}{2}$. The value of $x$ must be positive because it represents a side length. Therefore, $x=\frac{58\sqrt{2}}{2}$, or $x=29\sqrt{2}$. The perimeter, in inches, of the triangle is $58+x+x$, or $58+2x$. Substituting $29\sqrt{2}$ for $x$ in this expression gives a perimeter, in inches, of $58+229\sqrt{2}$, or $58+58\sqrt{2}$.

Choice A is incorrect. This is the length, in inches, of each of the congruent sides of the triangle, not the perimeter, in inches, of the triangle.

Choice B is incorrect. This is the sum of the lengths, in inches, of the congruent sides of the triangle, not the perimeter, in inches, of the triangle.

Choice D is incorrect and may result from conceptual or calculation errors.

### Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank